

Nigel Loh

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EDUCATION

University of Toronto Toronto, Canada
Bachelor of Science in Computer Science and Math (Co-op Program) (GPA: 3.97 / 4.0) *Expected June 2028*

- Coursework: Operating Systems, Machine Learning, Deep Learning, Data Structures, Algorithms, Databases, Computer Organization

WORK EXPERIENCE

Capital One McLean, Virginia
Incoming Software Engineer Intern

PointClickCare | *Python, MySQL, HuggingFace, Azure, OpenAI, LangChain* Toronto, Ontario
Machine Learning/Data Engineer Intern *May to August 2025*

- Designed and deployed NLP pipelines using **HuggingFace Transformers** to embed 10M+ unstructured clinical notes into dense vectors, enabling **semantic similarity search** and improving document retrieval speed by **60%**
- Built **RAG** pipelines with **LangChain**, and **OpenAI** to extract insights from over **5M clinical assessment questions**, reducing manual chart review time by **70%**
- Engineered high throughput data pipelines with **Pandas**, **NumPy**, and **MySQL** to ingest and preprocess **20M+ clinical records**, cutting ETL processing time by **40%**

PROJECTS

PokerFish | *React.js, FastAPI, Docker, AWS, NGINX, PostgreSQL, Redis, PyTorch*

- Deployed a full stack online **poker platform** on **AWS EC2** with **NGINX** as reverse proxy, optimizing traffic flow, security, and latency
- Implemented real time multiplayer gameplay using **WebSockets** with **Redis** for state synchronization, **React** frontend, and **FastAPI** backend, persisting game data in **PostgreSQL**
- Integrated an adaptive AI opponent trained with **Proximal Policy Optimization** in **PyTorch**, enabling dynamic user challenges

Dance Dance Revolution Keyboard (UofTHacks Winner) | *C++, Python, Arduino*

- Led a team of four to build a Dance Dance Revolution inspired keyboard designed to combat sedentary desk work by transforming typing into an active, movement based interaction
- Developed the hardware and firmware using **C++** and **Arduino**, with **Python** used to emulate standard keyboard inputs for seamless computer interaction
- Won **Best Hardware** at **UofTHacks** among **500+ participants** by creating an interactive input device that combines movement with productivity

MammoNet | *Python, PyTorch, Django, Next.js*

- Led a team at a **bioinformatics hackathon** to build a breast cancer classification system using a convolutional neural network, achieving **96 percent test accuracy** on microscopy images and placing in the top tier of submissions
- Developed scalable **REST APIs** with **Django** to handle image uploads, automated preprocessing, and real time model inference with minimal latency
- Connected the backend to a responsive **Next.js** interface, enabling instant predictions and dynamic **heatmap visualizations** to highlight malignancy indicators and assist clinical interpretation

TECHNICAL SKILLS

- Languages:** Java, Python, C, C++, TypeScript/JavaScript, Assembly, SQL
- Frameworks & Libraries:** React, Node.js, Next.js, PyTorch, TensorFlow, NumPy, Pandas, Scikit Learn, FastAPI
- Developer Tools:** Neovim, Git/Github, Linux/Unix, Docker, Firebase, NextAuth, OAuth 2.0, Bash, Zsh